ANSWER KEY & MARKING SCHEME SET 1

Chapter 12: **Magnetic Effects of Electric Current** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** Both A and R are true and R is the correct explanation of A.	At a crossing there would be two directions of the field.
2	**(c)** 90 degrees	Perpendicular is the maximum.
3	**(d)** 100 Hz	Proportional to speed.
4	**(b)** Both A and R are true but R is not the correct explanation of A.	Both true; the deflection is evidence of the field, not its explanation.
5	**(d)** zero	No change of linked field.
6	**(d)** 3 E	Proportional to rate of change.
7	**(a)** Both A and R are true and R is the correct explanation of A.	Needed for a temporary magnet.
8	**(b)** develops an induced current while the magnet moves	A changing field links the coil only during the motion.
9	**(b)** zero	Positive and negative halves cancel.
10	**(c)** A is true but R is false.	A is false: the force is zero when the conductor is parallel to the field; R is true.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Force is perpendicular to the velocity (1); No work is done, so the speed is constant (1).	1+1
12	I = 2200 / 220 = 10 A (1); The 15 A circuit is suitable (1).	1+1
13	It disconnects the appliance from the high potential (1); The appliance stays safe when switched off (1).	1+1
14	T = 1 / 60 = 0.0167 s (1); Reversals = 120 per second (vs 100) (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Force arises from the component of the field perpendicular to the wire (1); It is maximum when the wire is perpendicular (1); It is zero when the wire is parallel (1).	1+1+1
16	Metal case can become live due to a fault (1); The earth wire is a low-resistance path (1); The fuse blows and the user is safe (1).	1+1+1
17	Current produces a magnetic field (Oersted) (1); Changing magnetic field produces current (Faraday) (1); Force on a conductor in a field links the two (1).	1+1+1
	OR: Inside: field lines are nearly parallel and in one direction (1); Outside: lines go back in the opposite direction (1); Each element of the wire adds to the field (1).	1+1+1
18	Straight wire: concentric closed circles around the wire (1); Bar magnet: lines emerge from N and enter S (1); Both have closed loop fields (1).	1+1+1
	OR: Increase the number of turns (1); Use a soft iron core (1); Use a thicker core or a larger current (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Fleming's left-hand rule (1); Forefinger north, middle finger east (1); Thumb points upward (1); (a) Force becomes downward (1); (b) Both reversed: force upward as before (1).	1+1+1+1+1
	OR: Both have similar field patterns (1); Lines go from N to S outside (1); The solenoid's field can be switched on or off (1); A soft iron core makes it a strong electromagnet (1); Scrap is lifted while the current flows and dropped when switched off (1).	1+1+1+1+1
20	Armature coil rotates between magnet poles (1); EMI induces a current (1); Split rings reverse the connection every half rotation (1); Output is in one direction (1); Value varies between zero and maximum, so it is not steady (1).	1+1+1+1+1

	OR: Deflection increases with speed (1); It is proportional to the speed here (2 : 4 : 6 = 1 : 2 : 3) (1); The induced emf depends on the rate of change of the field (1); With double the turns the emf doubles (1); Deflections would be about 4, 8 and 12 divisions (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

ANSWER KEY & MARKING SCHEME SET 2

Chapter 12: **Magnetic Effects of Electric Current** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** clockwise	Clockwise appears as a south pole.
2	**(d)** 6000	100 per s x 60.
3	**(a)** Both A and R are true and R is the correct explanation of A.	Nothing is cut.
4	**(b)** anticlockwise when seen from above	Fingers curl anticlockwise from above.
5	**(d)** blows because I = 6.8 A	I = 1500 / 220 = 6.8 A > 5 A.
6	**(a)** Both A and R are true and R is the correct explanation of A.	Lower loss explains the preference.
7	**(a)** zero	No force when the motion is parallel to the field.
8	**(a)** Both A and B on together	1.5 kW draws 6.8 A.
9	**(a)** Both A and R are true and R is the correct explanation of A.	Prevents shock.
10	**(c)** moves in a straight line	No force along the field.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Induced emf is larger (1); More turns link more of the changing field (1).	1+1
12	Rectifier (1); Split-ring commutator (1).	1+1
13	Bring a compass near the wire (1); Deflection of the needle shows a current (1).	1+1
14	Iron is a magnetic material (1); Copper is not magnetised by an external magnet (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Total power = 3 kW (1); I = 3000 / 220 = 13.6 A (1); More than 5 A, so the fuse blows (1).	1+1+1
16	Bring the needle near an end (1); If the north pole of the needle is attracted, the end is a south pole (1); Reversing the current reverses the polarity (1).	1+1+1
17	The two sides of the coil exchange positions every half rotation (1); The direction of the induced current reverses each half rotation (1); Hence the output is alternating (1).	1+1+1
	OR: Move the magnet slowly: small deflection (1); Move it quickly: larger deflection (1); The rate of change of the field decides the emf (1).	1+1+1
18	Charge moves perpendicular to the field (1); Force is zero when it moves along the field (1); Maximum at 90 degrees (1).	1+1+1
	OR: The induced emf doubles (1); The frequency is unchanged (1); More turns link more field lines (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Each turn produces a magnetic field (1); Fields of all turns add inside (1); Field is stronger with more turns per unit length (1); Increasing the current raises the field (1); A soft iron core increases it considerably (1).	1+1+1+1+1
	OR: Frequency 50 Hz, period 0.02 s (1); Voltage 220 V (1); P = V I = 1100 W (1); E = 1.1 kW x 3 h (1); E = 3.3 kWh (1).	1+1+1+1+1
20	(a) Place a compass near the solenoid; switch on the current (1); Needle deflects, so current produces a magnetic field (1); (b) Connect the solenoid to the galvanometer (1); Move the magnet in and out (1); Deflection shows induced current from a changing field (1).	1+1+1+1+1
	OR: (a) Metal body may stay live, causing a shock (1); (b) Live wire remains connected when the fuse melts (1); (b) The appliance remains at high potential (1); (c) Appliance stays live even when switched off (1); Hence the fuse and switch must be in the live wire (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

ANSWER KEY & MARKING SCHEME SET 3

Chapter 12: **Magnetic Effects of Electric Current** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** At high potential with respect to earth	220 V.
2	**(a)** Both A and R are true and R is the correct explanation of A.	Breaking the live wire isolates the appliance.
3	**(a)** A stationary charge	Only moving charges and magnets.
4	**(a)** the fuse blows and the person is safe	Large current passes to earth.
5	**(a)** Both A and R are true and R is the correct explanation of A.	Force perpendicular to motion does no work.
6	**(b)** loses most of the magnetism if it is soft iron	Soft iron.
7	**(c)** decreases	Core enhances the field.
8	**(a)** Both A and R are true and R is the correct explanation of A.	The split ring acts as a commutator.
9	**(c)** decreases	Turns per unit length are halved.
10	**(d)** cuts the lines of force at the greatest rate	Greatest rate of change of flux.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Field is proportional to the number of turns per unit length (1); B has double the field (1).	1+1
12	It acts like a bar magnet (1); It aligns with the earth's field (1).	1+1
13	Field is strongest near the poles (1); Field lines are closer there (1).	1+1
14	Each piece has its own north and south poles (1); Isolated poles are not possible (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	One cycle for one rotation (1); Frequency = 50 Hz (1); Period = 0.02 s (1).	1+1+1
16	Deflection occurs in one direction while entering (1); In the opposite direction when leaving (1); The field through the coil first increases and then decreases (1).	1+1+1
17	The needle aligns with the magnetic field (1); Without a field it points in no fixed direction (1); It is used to find direction by the earth's field (1).	1+1+1
	OR: Metal body connected to the earth through the earth wire (1); If the live wire touches the body, current passes to earth (1); Prevents shocks and fires (1).	1+1+1
18	(a) To take leakage current to earth (1); (b) The live wire carries the full current, and cutting it isolates the appliance (1); (c) Thick wire has low resistance and less heating (1).	1+1+1
	OR: The appliance is cut off from the circuit (1); The wire may still be live, so it is dangerous (1); Switch and fuse give protection (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Live, neutral and earth wires; appliances in parallel (1); Fuse and MCB protect from overload (1); Earth wire gives safety (1); Total power = 300 + 160 + 1500 = 1960 W (1); I = 1960 / 220 = 8.9 A (1).	1+1+1+1+1
	OR: Oersted: current produces a magnetic field (1); Led to the electromagnet (1); Faraday: a changing field induces a current (1); Led to the generator (1); Both show the link between electricity and magnetism (1).	1+1+1+1+1
20	Depends on current (1); On the strength of the field (1); On the angle between wire and field (1); Reverse the current or field to reverse the force (1); Field up and current right (both along the page): force is out of the page, towards the observer (1).	1+1+1+1+1

	OR: While pushing in: the needle deflects one way (1); While held at rest: no deflection (1); While pulling out: deflection in the opposite direction (1); Induced current appears only when the field linked with the coil changes (1); Faster motion gives a larger deflection (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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